Karmanya Batra

New Delhi, India | +91 9971521981 | karmanya82@gmail.com | linkedin.com/in/karmanya-batra | github.com/KarmanyaBatra

Professional Summary

Full Stack Developer and AI/ML Engineer with hands-on experience across web development, IoT, and computer vision, including real-world defense-focused projects for the Indian Army. Skilled in React.js, Next.js, Node.js, Express.js, MongoDB, PostgreSQL, Python, and RESTful API design, with a track record of building production-ready applications spanning smart automation, computer vision, and chatbot-integrated systems. Delivered measurable improvements of 20%+ in speed, reliability, and user experience across multiple full-stack and ML projects.

Education

Vellore Institute of Technology — Bhopal, India
B.Tech in Computer Science & Engineering (AI & ML) — *2022 – 2026*

- **CGPA / GPA: 7.7/10.0 (3.85/5.0)**
- Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, DBMS, Operating Systems, Software Engineering, Object-Oriented Programming, Computer Networks

Technical Skills

Languages: Python, JavaScript (ES6+), C++, Java, SQL
Frontend & Backend: React.js, Next.js, HTML5, CSS3, Node.js, Express.js, Flask, RESTful APIs, JWT, bcrypt, React Router, React Hooks
Databases: MongoDB, PostgreSQL, MySQL, SQLite, Mongoose ODM, Prisma ORM, Database Indexing, Query Optimization
ML & AI: TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV
Cloud & DevTools: AWS, Git, GitHub, Postman, Docker (basic), Jupyter Notebook, VS Code, Raspberry Pi
Core Competencies: CNN & Transfer Learning, Data Augmentation, Cross-Validation, Agile Development, CI/CD Basics

Experience

Frontend and AI Intern — May 2026 – Present
Rayonix — *Gurugram, India*

- Built a real-time LED detection and calibration pipeline for an Indian Army surveillance application using Python and OpenCV with ArUco marker mapping, improving detection accuracy by **12%**.
- Deployed the LED detection pipeline on Raspberry Pi 4 for edge inference, cutting average detection latency by **30%**.
- Developed Next.js and React.js frontend interfaces for a smart switch automation platform, enabling remote control of **15**+ IoT devices.
- Optimized **8**+ REST API endpoints and PostgreSQL schemas via Prisma ORM, reducing average query response time by **18%**.
- Shipped **10**+ production features using Git-based feature branches and pull requests within sprint deadlines.

Software Developer Intern — May 2025 – July 2025
Besseggen Infotech LLP — *Remote, India*

- Decreased UI load time by **20%** by implementing React.js lazy loading and code-splitting across **10**+ frontend components.
- Streamlined **3**+ manual workflows by integrating Node.js and Express.js RESTful APIs, cutting team effort by **30%**.
- Resolved **15**+ critical production bugs using Postman and monitoring pipelines, reducing resolution time by **40%**.
- Redesigned **5**+ REST API endpoints with the backend team, improving throughput and reliability by **22%**.

Technical Projects

Attendance Management System with Chatbot | *React, Flask, SQLite* — 2025

- Built a React, Flask, and SQLite attendance system supporting **500**+ student records and **50**+ concurrent users.
- Integrated an NLP-based chatbot handling **10**+ query types, cutting manual attendance lookups by **40%**.
- Designed **6**+ REST API endpoints for attendance marking, retrieval, and chatbot query processing.
- Added input validation and error handling, reducing malformed-query failures by **25%**.

Alzheimer's Disease Classifier | *TensorFlow, Keras, Python* — Apr 2025 – May 2025

- Benchmarked **3** CNN architectures (VGG16, ResNet50, EfficientNet), achieving **92%** validation accuracy across **4** categories.
- Improved model generalization by **15%** via transfer learning and data augmentation.
- Cut training time by **35%** using GPU-optimized batch processing pipelines.
- Applied Grad-CAM visualization across **4** disease classes to improve model interpretability.

Achievements

Featured Speaker | *BBC World Service, What in the World Podcast* — April 2026

- Spoke on graduate employment challenges; episode reached **19,000**+ viewers on YouTube.

Event Organiser | *iOS Club Inauguration, VIT Bhopal* — Dec 2022

- Collected **100**+ feedback responses to inform future club activities.